

Missing Roots and Sign Errors in Quadratic Solutions

- Two errors account for most lost marks on quadratic equations: reporting an **incomplete solution set** (one root when there are two), and a **sign slip** in the quadratic formula.
 - Unless a problem says otherwise, “solve” means **list every solution**.
 - Where a worked attempt is shown, the attempt contains exactly one mistake. Find it before you start computing.
 - Substituting an answer back into the original equation is the check that catches both error types — use it.
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1. Solve, and list every solution: $x^2 = 49$

Answer: _____

2. Dana solves $2x^2 - 32 = 0$ like this: “ $2x^2 = 32$, so $x^2 = 16$, so $x = 4$.”

- (a) Name the mistake in Dana’s work.
- (b) Write the complete solution set. Put it on the answer line.

Answer: _____

- 3.** Solve, and list every solution: $x^2 - 5x = 0$

Answer: _____

- 4.** Marco solves $x^2 = 6x$ by dividing both sides by x , which gives him $x = 6$.

- (a) Name the error in Marco's method, and say why it costs him a solution.
 (b) Write the complete solution set. Put it on the answer line.

Answer: _____

- 5.** Solve by factoring, and list every solution: $x^2 + 2x - 15 = 0$

Answer: _____

6. Solve with the quadratic formula, and list every solution: $x^2 - 6x + 5 = 0$

Write down the values of a , b and $-b$ before you substitute.

Answer: _____

7. Use the quadratic formula on $2x^2 - 7x + 2 = 0$. Give the *larger* of the two solutions, rounded to two decimal places.

Answer: _____

8. Nadia solves $2x^2 + 5x - 3 = 0$ with the quadratic formula. Her substitution under the square root is correct, but she writes the denominator as 2 instead of $2a$, and reports $x = 1$ and $x = -6$.

- (a) Name the error in Nadia's work.
- (b) Write the correct solution set. Put it on the answer line.
- (c) Explain how substituting one of Nadia's two answers into the original equation would have shown her that something was wrong.

Answer: _____